

# TWO CLOSURES AND A SURVIVOR: THE AMPLIFICATION ROUTE TO THE RIEMANN HYPOTHESIS THROUGH ALMOST-PRIME SIEVE BOUNDS

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ABSTRACT. In a companion paper [8] we argued that every attempted proof of the Riemann Hypothesis (RH) within the Connes–Consani–Moscovici (CCM) framework presents RH as the boundary point of a one-parameter family whose interior is unconditionally controlled, that no one-dimensional propagation principle reaches that boundary, and that exactly two logical forms survive the resulting no-go: *two-dimensional propagation* (Form A) and *index amplification* (Form B), with a cheap auxiliary line (Form C, an effective version of Bagchi’s self-approximation criterion). We now report the outcomes. *Form C closes*: the self-approximation density does not degenerate as the comparison parameter  $d \rightarrow 0$ ; the scale degenerates linearly, and the underlying amplifier is self-blocking—the anchored density that would force RH and the zero-density ceiling that bounds it under  $\neg$ RH carry the same exponent, so any improvement to one improves the other. *Form A closes*: the only joint heat-time/arithmetical trace  $\mathcal{T}(t, x)$  whose two edges are unconditional theorems vanishes identically in the interior, because the forward De Bruijn–Newman flow preserves reality of zeros; every coupling inequality is vacuous. *Form B survives*. We prove that the negative index is strictly superadditive under the Krein tensor product,  $\text{neg. ind}(Q_1 \otimes Q_2) = p_1 q_2 + q_1 p_2$ , giving unconditional exponential amplification  $\kappa(\psi^{\otimes k}) = (4m)^k/2$  on the off-critical subspace  $\mathcal{K}_{\text{off}}$ ; and we execute the decisive experiment of [8, 7]: the two-variable Weil functional restricted to a thickened multiplicative diagonal  $h(x_1, x_2) = g(x_1 x_2) \varphi(x_1/x_2)$  has (i) a spectral side realising the Minkowski sum of zero pairs with the predicted  $(8, 8)$  signature, and (ii) an *autonomous* arithmetic side supported on almost-primes,  $\sum_N g(N) \sum_{d|N} \Lambda(d) \Lambda(N/d) \varphi(d^2/N)$ —a weighted transform of  $\Lambda \star \Lambda$  computable without knowledge of the zeros. The Hadamard barrier does not reappear inside the functional. The entire remaining difficulty concentrates in one statement, the *sieve-to-index bridge*: that an unconditional sieve upper bound on the almost-prime functional controls the negative-index growth of the tensor powers. We state it precisely, explain why the parity phenomenon does not obstruct it (one must *bound*, not *distinguish*), and identify the wall it would meet if it required asymptotics rather than bounds—the pair-correlation wall, which is *not* a positivity wall. No proof of RH is claimed.

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## 1. INTRODUCTION

**1.1. Where we stand.** The companion paper [8] reduced the search for a proof of RH, within the CCM framework, to a structural trichotomy. After a programme that mapped seven mutually reinforcing walls (MW-1 through MW-7; see [8, 7]), we argued that the obstruction has a single logical form: RH is always the boundary value of a one-parameter family whose interior is an unconditional theorem, and no one-dimensional propagation principle (limit, analyticity, parabolic backward uniqueness, monotonicity, ergodic recurrence) carries the interior control to the boundary. By elimination, two argument-forms remain: two-dimensional propagation (Form A) and index amplification (Form B). A third, cheaper line—an effective version of the Bagchi self-approximation criterion (Form C)—was proposed as a low-risk probe.

This paper reports what happened when each was pursued. The arithmetic of the report is simple: Forms A and C close, with precise reasons that are themselves informative; Form B survives a first decisive experiment and concentrates the entire remaining difficulty into one unconditional lemma. The purpose of the paper is to record the two closures honestly and to state the surviving problem sharply enough that it can be attacked or refuted.

**1.2. What this paper proves, and does not.** We prove three closure/structure theorems (Theorems 2.1, 3.1, and the index calculations 4.1–5.1), all unconditional. We do *not* prove RH, nor any conditional statement that would follow from a weaker hypothesis than is already known. The contribution is the localisation of the difficulty: after this paper, the amplification route to RH rests on exactly one missing lemma (Missing Lemma 6.1), of a type—a sieve upper bound on a bilinear almost-prime functional—that lies in the one region where unconditional methods are known to operate.

**1.3. Framework.** We use the notation of [6, 7, 8]. Briefly:  $dm_\infty(s) = (2\pi)^{-2}|\Gamma(1/4 + is/2)|^2 ds$  is the archimedean spectral measure;  $d\nu = dm_{\text{full}} - dm_{\text{full, on}} \geq 0$  is the unconditionally non-negative discrepancy measure with RH  $\iff d\nu = 0$  [7, 89]; the CCM trace criterion holds at a single scale, RH  $\iff T_\lambda[\lambda_0] = 0$  [7, ]; and the bridge theorem [5] identifies the Weil quadratic form's negative index with the negative index of the operator  $H_C$  on the Pontryagin space  $(K, Q)$ ,  $\kappa(Q) = \text{neg. ind}(H_C) = 2m$ , localised on the off-critical subspace  $\mathcal{K}_{\text{off}}$  [7, Thm 8.1]. Here  $m$  is the number of off-critical zero quadruples; RH  $\iff m = 0$ .

## 2. FORM C CLOSES: THE SELF-BLOCKING AMPLIFIER

**2.1. The proposal and its mis-framing.** Bagchi's criterion makes RH equivalent to the strong recurrence of  $\zeta$  [7, ]. The self-approximation family  $\text{SR}_d$ —comparing  $\zeta(s+i\tau)$  to  $\zeta(s+id\tau)$  on a disk in the strip with positive lower density of admissible  $\tau$ —is proved unconditionally for every  $d \neq 0$  (Baker's theorem on linear forms in logarithms for algebraic irrational  $d$ ; extensions of Nakamura, Pańkowski, Garunkštis covering all real  $d \neq 0$ ), while  $d = 0$  is exactly RH. Form C asked: as  $d \rightarrow 0$ , how do the constants degrade? [8] posed a dichotomy—exponential degradation (quantified MW-2, closed) versus polynomial (a genuine tension worth exploiting).

The dichotomy was mis-framed, and seeing why is the content [7, ].

**Theorem 2.1** (The amplifier is self-blocking). *The lower density  $\delta(d, \varepsilon)$  of admissible shifts does not degenerate as  $d \rightarrow 0$ : there is  $d_0(\varepsilon) > 0$  with  $\delta(d, \varepsilon) \geq \delta^*(\varepsilon) > 0$  uniformly for  $0 < |d| < d_0(\varepsilon)$ . What degenerates is the scale: the first guaranteed shift satisfies  $T_0(d, \varepsilon) \asymp C(\varepsilon)/|d|$ , linearly. Moreover the Rouché amplification that would convert one off-critical zero into a positive-density family of off-critical zeros is bounded, under  $\neg\text{RH}$ , by an anchored density with the same exponent  $\kappa(\sigma)$  as the unconditional zero-density ceiling  $N(\sigma, T) = O(T^{\kappa(\sigma)})$  that contradicts it. The threshold density needed to force RH and the ceiling that caps it move together under any improvement of  $N(\sigma, T)$ .*

The proof [7, ] isolates a diophantine-free regime ( $|d| < d_0(\varepsilon)$ , where the truncated frequency system carries no linear relation and Baker is not invoked), shows the degradation is at worst polynomial in  $1/|d|$  where diophantine input does enter, and—decisively—shows the Rouché step consumes density not globally but in the *anchored window*  $J_d = \{\tau : d\tau \approx \gamma_0\}$ , where the slow copy is frozen and the statement collapses to universality against a *forbidden* target (a shifted  $\zeta$ , which has a zero in the disk): the admissible density there is provably zero by Hurwitz. The wall is binary, not a rate.

**Corollary 2.2.** *A mobile choice  $d = d(T) \rightarrow 0$  does not help: anchoring ( $|d|T = O(1)$ ) freezes the slow copy and collapses  $\text{SR}_{d(T)}$  to  $\text{SR}_0 = \text{RH}$ ; equidistribution ( $|d|T \rightarrow \infty$ ) destroys the anchor and yields at most  $o(T)$  replicated zeros. The two regimes are complementary and exhaustive [7, Lem. 6.1].*

*Observation 2.3.* Form C is a clean instance of MW-2 (the arithmetic propagation wall) made quantitative. The missing lemma that would close it (localised universality against a forbidden target in short intervals) is, on inspection, a re-encoding of RH itself. We record it as such and do not pursue it.

### 3. FORM A CLOSES: THE DEGENERATION THEOREM

**3.1. The proposal.** Form A sought a two-dimensional propagation principle. Let  $\mathcal{T}(t, x)$  be the CCM discrepancy trace of the finite-Euler approximant indexed by  $x$ , evolved by the De Bruijn–Newman (DBN) heat flow to time  $t$ . The proposal [8, Prob. 6.2] was that the two edges— $\mathcal{T} = 0$  for  $t \geq \Lambda$  (the DBN constant) and reality of the approximant’s zeros for finite  $x$ —together with a coupling inequality between  $\partial_t \mathcal{T}$  and  $\partial_x \mathcal{T}$  might propagate control to the corner  $(t, x) = (0, \infty) = \text{RH}$ , even though neither axis reaches it alone.

**3.2. The closure.** The coupling never gets a chance to exist.

**Theorem 3.1** (Degeneration). *The only definition of  $\mathcal{T}(t, x)$  for which both edges are unconditional theorems is one that vanishes identically on the interior of the quadrant. Consequently  $\partial_t \mathcal{T} \equiv 0$ ,  $\partial_x \mathcal{T} \equiv 0$ , and every parabolic, monotone, or log-convex coupling inequality is vacuous.*

*Reason.* The forward DBN heat flow preserves reality of zeros: if  $\Xi_t$  has only real zeros at one time it has only real zeros for all later times (de Bruijn 1950 [3]). But “reality of zeros” is precisely the property defining the finite- $x$  (arithmetic) edge. Hence the trace that is zero on the arithmetic edge is forced to be zero throughout the forward-invariant interior. Making both edges theorems and the interior non-trivial are incompatible [7, Thm 3.2].  $\square$

The corner  $(0, \infty)$  thus remains a pure discontinuity, whose removal is the one-dimensional continuity wall already identified in [8, §5.1]. Form A reduces to that wall.

**3.3. Two byproducts.** The analysis [7, ] produced two unconditional byproducts of independent interest. First, the only non-degenerate ledger—the Weil side at  $x = \infty$ —admits a closed form for the first kernel block: for  $N(\lambda_0) = 1$ ,

$$B_{\lambda_0}(r) = \frac{1}{8} \operatorname{sech}^{1/2}(r) (105 \operatorname{sech}^4 r - 78 \operatorname{sech}^2 r + 5),$$

whose per-prime increments are positive for prime powers  $p^m \leq 7$  and negative for  $p^m \geq 8$ , and whose prime series *diverges* like  $(5\sqrt{2}/8) \log X$ . Hence the trace identity of [7, Thm 4.1] is strictly formal: the CCM test function sits exactly on the boundary of the Weil class (the poles of the weight at  $\pm i/2$ ). Second, the analysis corrected two prior errors in the corpus: a sign in the evolution equation [7, ] and an omitted on-critical block [7, ], the latter invalidating an earlier sign heuristic and reopening the sign of  $\partial_t T|_0$  under  $\neg \text{RH}$  as a declared gap.

## 4. FORM B SURVIVES: AMPLIFICATION OF THE NEGATIVE INDEX

**4.1. The Deligne skeleton, transplanted.** Deligne’s proof over function fields [2] is, in skeleton, not a positivity argument: an integer invariant (the weight), an amplification (a tensor power multiplies the defect), a uniform sublinear bound (rationality: the poles do not move), and the conclusion that an integer bounded by something sublinear in its own multiple must vanish. The CCM programme already supplies the integer invariant:  $\kappa(Q) = 2m$ , localised on  $\mathcal{K}_{\text{off}}$ . Form B asks for the amplification.

**4.2. The negative index is strictly superadditive.** The first half of the falsifier stated in [8, §7]—“perhaps  $\kappa$  is exactly additive under every available product”—is false.

**Theorem 4.1** (Tensor index). *For finite-dimensional Krein spaces with quadratic forms of signature  $(p_1, q_1)$  and  $(p_2, q_2)$ ,*

$$\text{neg. ind}(Q_1 \otimes Q_2) = p_1 q_2 + q_1 p_2.$$

*This is strictly superadditive in the negative index whenever  $(p_1 - 1)q_2 + (p_2 - 1)q_1 > 0$ .*

*Proof.* In a Sylvester basis diagonalising each  $Q_i$  to  $\pm 1$ , the product basis diagonalises  $Q_1 \otimes Q_2$ ; a product vector is negative iff exactly one factor is negative, giving  $p_1 q_2 + q_1 p_2$  negative directions.  $\square$

**Theorem 4.2** (Exponential amplification on  $\mathcal{K}_{\text{off}}$ ). *With  $m \geq 1$  off-critical quadruples,  $\psi = (\mathcal{K}_{\text{off}}, Q|_{\mathcal{K}_{\text{off}}})$  has signature  $(2m, 2m)$ , and under the Krein tensor product*

$$\kappa(\psi^{\otimes k}) = \frac{1}{2}(4m)^k \geq k \cdot 2m = k \kappa(\psi) \quad (k \geq 1),$$

*with equality only at  $k = 1$ . Moreover  $m = 0 \Rightarrow \kappa(\psi^{\otimes k}) = 0$  for all  $k$ : the invariant of the powers detects RH at every level.*

This is ingredient (1)–(2) of the skeleton, unconditional. The full tensor product on  $K$  (not its restriction to  $\mathcal{K}_{\text{off}}$ ) is infinite-dimensional and gives  $\kappa = \infty$  already at  $k = 2$ ; the legitimate, finite invariant is the *relative* index on  $\mathcal{K}_{\text{off}}$  [7, §3].

**4.3. The sterility lemma.** Why is this not already a proof? Because amplification alone is sterile.

**Proposition 4.3** (Sterility). *In the abstract Krein-tensor setting, both the amplification (a) and any bound on  $\kappa(\psi^{\otimes k})$  (b) are computed from the same signature data. No contradiction can be extracted from (a) and (b) together unless (b) is supplied by a source independent of the signature.*

In Deligne the independent source is cohomological rationality: the poles of the  $L$ -function of the tensor family lie at prescribed places, with an additive error uniform in  $k$ , guaranteed by the finite-dimensionality of the  $\ell$ -adic cohomology. The question for Form B is whether the CCM framework possesses an analogous independent finiteness source. It does not possess it for free—the tensor product must first be *realised* analytically, as a functional with its own explicit formula—and this is the experiment we now execute.

## 5. THE DECISIVE EXPERIMENT: TWO-VARIABLE WEIL WITH A DIAGONAL KERNEL

**5.1. Setup.** We must realise the tensor square  $(\mathcal{K}_{\text{off}}, Q|_{\mathcal{K}_{\text{off}}})^{\otimes 2}$  as a spectral block of a functional whose arithmetic side is computable without the positions of the zeros (Missing Lemma 5.2 of [7, ]). The natural candidate is the two-variable Weil functional. Applied to a product kernel  $h(x_1, x_2) = f_1(x_1)f_2(x_2)$  it factors as  $\mathcal{W} \otimes \mathcal{W}$ , with zero term  $\sum_{\rho_1, \rho_2} \hat{f}_1(\rho_1)\hat{f}_2(\rho_2)$ . The eigenvalues of  $H_C \otimes I + I \otimes H_C$  are the sums  $\rho_1 + \rho_2$ ; to see them we must *couple* the variables.

**5.2. The thickened diagonal.** The naive diagonal  $h(x_1, x_2) = g(x_1 x_2)$  diverges: with  $u = x_1 x_2$ ,  $v = x_1$ , the inner  $v$ -integral  $\int_0^\infty v^{\rho_1 - \rho_2 - 1} dv$  has no decay. The well-defined object is the *thickened* multiplicative diagonal

$$(1) \quad h(x_1, x_2) = g(x_1 x_2) \varphi(x_1/x_2), \quad \varphi \in C_c^\infty(\mathbb{R}_+^*).$$

**Theorem 5.1** (Spectral side; the Minkowski sum). *With  $u = x_1 x_2$ ,  $v = x_1/x_2$  (Jacobian  $dx_1 dx_2 = du dv/2v$ , exponent  $v^{(\rho_1 - \rho_2)/2 - 1}$ ), the Mellin transform of (1) is*

$$\hat{h}(\rho_1, \rho_2) = \frac{1}{2} \hat{g}\left(\frac{\rho_1 + \rho_2}{2}\right) \hat{\varphi}\left(\frac{\rho_1 - \rho_2}{2}\right),$$

so the zero term becomes  $\sum_{\rho_1, \rho_2} \hat{g}\left(\frac{\rho_1 + \rho_2}{2}\right) \hat{\varphi}\left(\frac{\rho_1 - \rho_2}{2}\right)$ : the Minkowski sum of zero pairs, weighted in their difference. For one quadruple  $\rho \in \{\frac{1}{2} \pm \delta \pm i\gamma_0\}$ , the sixteen pairs  $(\rho_i, \rho_j)$  organise under  $\rho \mapsto 1 - \bar{\rho}$  into eight fixed-point-free orbits—eight hyperbolic planes—so the 16-dimensional block has signature exactly  $(8, 8)$ , matching  $\text{neg. ind}(Q|_{\mathcal{K}_{\text{off}}^2}) = 8$  of Theorem 4.1, with no degeneration.

*Remark 5.2* ( $\varphi$  carries the index). A finding not anticipated in [8, 7]: the thickening weight  $\hat{\varphi}$  is not a mere regulariser. Of the eight negative directions, only two sit at the real sums  $\rho_i + \rho_j - 1 = \pm 2\delta$ ; the other six live at purely imaginary sums, where the off-criticality  $\text{Re} = \frac{1}{2} \pm \delta$  is registered *only* by  $\hat{\varphi}$  through  $\text{Re}\left(\frac{\rho_i - \rho_j}{2}\right) = \pm \delta$ . The difference variable is a genuine carrier of the negative index, not an artefact of regularisation.

**5.3. The arithmetic side is autonomous.** This is the decisive point. The Hadamard barrier MW-7 would reappear if the mixed zeros $\times$ primes terms of the doubled functional required the positions of the zeros. They do not.

**Theorem 5.3** (Mixed terms reabsorb). *Applied slice by slice—fixing the first variable at a prime power  $n$  and using the one-variable explicit formula on  $x_2 \mapsto h(n, x_2) \in C_c^\infty$ —the apparent zeros $\times$ primes terms of the two-variable functional collapse to arithmetic $\times$ arithmetic terms. MW-7 does not appear inside  $\mathcal{W}_2$  [7, Thm 4.4].*

**Theorem 5.4** (Almost-prime functional). *For the thickened diagonal (1), the primes $\times$ primes block of  $\mathcal{W}_2$  is*

$$(2) \quad \sum_{N \geq 1} g(N) \sum_{d|N} \Lambda(d) \Lambda(N/d) \varphi(d^2/N),$$

a  $\varphi$ -weighted transform of the Dirichlet convolution  $\Lambda \star \Lambda$ —the almost-prime part of Selberg's  $\Lambda_2(N) = \Lambda(N) \log N + (\Lambda \star \Lambda)(N)$ —together with three finitely-supported terms with  $g, \varphi$  interchanged. It is computable from the primes alone, without the positions of the zeros [7, Thm 4.2].

Theorems 5.1–5.4 together verify the two conditions of Missing Lemma 5.2 of [7, ]: the doubled Weil functional  $\mathcal{W}_2$  realises the tensor-square spectral block, and its arithmetic side is autonomous, living on the almost-prime semigroup. The candidate for the analytic realisation  $X_2$  is thus  $\mathcal{W}_2$  restricted to thickened diagonals, with  $Q_2(h, h') = \mathcal{W}_2(h \star \tilde{h}')$ .

## 6. WHAT WE WANT TO PROVE: THE SIEVE-TO-INDEX BRIDGE

**6.1. The honest accounting.** The amplification (Theorem 4.2) and the realisation (§5) are in hand. By the sterility lemma (Proposition 4.3), the contradiction requires an *independent* bound on the negative-index growth of the tensor powers. The arithmetic side (2) is exactly such a potential source: it is a functional on almost-primes, supplied by the primes, not by the zeros. The remaining problem is to convert a bound on this functional into a bound on the index.

**Missing Lemma 6.1** (The sieve-to-index bridge). Let  $X_k$  be the  $k$ -fold thickened-diagonal Weil functional realising  $(\mathcal{K}_{\text{off}}, Q|_{\mathcal{K}_{\text{off}}})^{\otimes k}$ , with arithmetic side the order- $k$  almost-prime functional generalising (2). Prove an unconditional upper bound of *sieve type*

$$\kappa(Q_k^{\text{rel}}) \leq f(k), \quad f(k)/k \rightarrow 0,$$

on the relative negative index of  $Q_k$  on the off-critical block, with  $f(k)$  controlled by the almost-prime functional—the analogue of “the poles do not move,” here a uniform sieve bound rather than a positivity.

If proved, Missing Lemma 6.1 closes the argument: combined with the unconditional lower bound  $\kappa(\psi^{\otimes k}) \geq 2mk$  (Theorem 4.2, in its robust sublinear form), it forces  $2mk \leq f(k)$  for all  $k$ , hence  $m = 0$ , i.e. RH. We claim no proof of it.

**6.2. Why parity does not obstruct it.** The almost-prime location of (2) is not an accident, and it is the reason for cautious optimism. The parity phenomenon—the obstruction that closed the  $\omega$ -class programme (Arc A of [7]) and that limits all sieve methods—forbids *distinguishing*  $\omega(n) \bmod 2$ , equivalently forbids sieves from producing *asymptotics* for sequences detected by sign. But Missing Lemma 6.1 asks for an *upper bound* on a bilinear almost-prime functional, not an asymptotic and not a sign discrimination. Upper bounds on  $\Lambda \star \Lambda$ -type sums over almost-primes are exactly what the Selberg sieve, Vaughan’s identity, and the Heath-Brown decomposition deliver unconditionally. The functional (2) lives in the one region where the methods operate.

*Remark 6.2* (The wall it would meet, if any). We state the risk plainly. If the bridge required not an upper bound but an *asymptotic* evaluation of (2)—if the index growth were governed by the main term of the almost-prime functional rather than by a majorant—then the relevant obstruction is the pair-correlation/twin-type barrier (Montgomery–Goldston), the difficulty of evaluating  $\sum \Lambda(n)\Lambda(n+h)$ -type correlations. This is a real wall, but it is *not* the positivity wall MW-1/MW-4: it is an evaluation barrier, not a sign barrier. No route in the programme has previously terminated against it. Whether Missing Lemma 6.1 needs a bound (passable) or an asymptotic (the pair-correlation wall) is the first question to settle, and it is a finite analytic question about (2).

**6.3. The shape of a proof, and its falsifiers.** A proof of Missing Lemma 6.1 would proceed in three steps, each falsifiable:

- (1) *Index from functional.* Express  $\kappa(Q_k^{\text{rel}})$  as the count of negative eigenvalues of the bilinear form  $Q_k(h, h') = X_k(h \star \tilde{h}')$  on the thickened-diagonal test space, and bound it by the number of sign changes of the arithmetic kernel (2). *Falsifier:* if the negative index is not controlled by the kernel’s sign changes (e.g. if the archimedean block dominates), the arithmetic side is irrelevant and the route is dead.
- (2) *Sieve majorant.* Bound the almost-prime functional by a Selberg majorant uniform in  $k$ . *Falsifier:* if the only available bound grows like  $k$  or faster (not  $o(k)$ ), the sterility lemma is not overcome.
- (3) *Uniformity.* Verify  $f(k)/k \rightarrow 0$  with the implied constant independent of  $m$ . *Falsifier:* if uniformity in  $m$  fails, the bound cannot kill all  $m \geq 1$  simultaneously.

Step (1) is the immediate next target; it is finite and self-contained.

## 7. ASSESSMENT

We have closed two of the three surviving forms, with reasons that sharpen the map: Form C is MW-2 quantified into a self-blocking amplifier (Theorem 2.1); Form A is a degeneration forced by the forward heat flow (Theorem 3.1). The survivor, Form B, has cleared its first decisive test: the negative index amplifies exponentially and unconditionally (Theorem 4.2), and the tensor square is realised by a two-variable Weil functional whose arithmetic side is autonomous and supported on almost-primes (Theorems 5.1–5.4), with the Hadamard barrier absent from the functional.

The entire weight now rests on one unconditional statement, Missing Lemma 6.1, of a kind that lives where sieve methods work, and whose only foreseeable wall is an evaluation barrier distinct from every positivity wall the programme has met. We make no prediction about whether it is provable. We claim only that the search for the missing idea has narrowed to a single, finite, falsifiable analytic question: does an unconditional sieve bound on the almost-prime functional (2) control the negative-index growth of its tensor powers? That narrowing—not any claim of progress toward RH itself—is the contribution of this paper.

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